SI Session Planning Form

SI Leader: Brandon Cobb
Session Week: 15
Session#: 15
Instructor: Professor Tramontozzi
Old Plans: SharePoint
Session Goal: Students will review and apply concepts of concentration and solution chemistry, including calculating molarity, mass percent, volume percent, and mass/volume percent. They will apply dilution principles and use solubility rules to determine whether a solution is unsaturated, saturated, or supersaturated, and to predict precipitation. Students will analyze factors affecting solubility, including temperature and the polarity of solute and solvent. They will also explore colligative properties such as freezing point depression and boiling point elevation for electrolytes and nonelectrolytes. Additionally, students will practice stoichiometric calculations, identify limiting reactants and theoretical yields, and classify chemical reactions including synthesis, decomposition, single and double replacement, combustion, neutralization, and redox reactions. By the end of this unit, students will have mastered understanding the difference between dissociation and ionization, calculating pH, pOH, $[H_3O^+]$, and $[OH^-]$, determining percent ionization for weak acids, interpreting K_a and pKa values, identifying conjugate acid-base pairs, analyzing diprotic and triprotic acids, and predicting how salts affect the pH of a solution through hydrolysis. They will also be able to apply these concepts to assess whether solutions are acidic, basic, or neutral, and to check the consistency of their calculations with proper units and significant figures.

Opener

Take Attendance

Collaborative Learning Techniques	Turn to a Partner
Learning Strategy	Ice Breaker (Two truths and a Lie)
Time	11:30—11:35
Materials	Notes, homework, whiteboard, markers, ready to speak
Lecture/Textbook/Old Plans/Slide References	Class roster

Definitions

1. **Turn to a Partner:** Group members work with a partner on an assignment or discussion topic. This technique works best with group participants who have already been provided with enough background on a subject that they can immediately move to a discussion with their partner without previewing or reviewing concepts.

2. **Ice Breaker:** Each student states two true facts and one false statement related to chemistry concepts (or general academic experiences), and the group guesses which statement is false. This activity encourages peer interaction, sparks discussion, and lightly reviews content in a fun, low-pressure way.

Checking for Understanding (questions))

- What is the primary goal of this activity?
- How do students know if their solution at a station is correct? .
- What should students do if they are unsure about a question?
- How does moving between stations benefit student learning?
- What is the importance of showing work on the team slip?
- How are bonus points awarded during the activity? n.
- What is the role of the facilitator at each station?
- How should teams handle disagreements while solving a question?
- Why is it important that only verified answers are recorded on the team slip?
- What is the wrap-up discussion intended to accomplish?

Activity 1

Collaborative Learning Techniques	Group Discussion
Learning Strategy	Lecture Review
Time	11:40—12:05
Materials	Notes, homework, whiteboard, markers, ready to speak, periodic table
Lecture/Textbook/Old Plans/Slide References	Chapter 8: Solutions, Chapter 9: Chemical Reactions

Definitions

1. **Group Discussion:** One person in the group is asked to present on a topic or review material for the group and then lead the discussion for the group. This person should not be the regular group leader.
2. **Lecture Review:** As a group summarize the lecture from the previous class. You may have to provide prompts for the students. For example, "The first concept discussed was Civil Liberties and Public Policy, what did the profressor highlight regarding this?" You may want to ask them to try summarizing without looking at their notes; however, if they are having a difficult time remembering, tell them to refer to their notes.

Activity Overview

1. **Purpose:** Provide students with a dynamic, interactive review session that encourages collaboration, reasoning, and problem-solving across multiple concept areas. The activity simulates a "challenge pathway" in which teams work through multiple stations to reinforce mastery of previously taught material.
2. **Structure:**
   - Students work in pairs or small teams.
   - The classroom is organized into multiple stations, each focused on a specific conceptual area (e.g., calculations, prediction exercises, conceptual reasoning).
   - Each station contains laminated question cards with problems, a space for showing work, and instructions for submitting answers.
   - Teams choose a station to start and move freely between stations after completing questions.
3. **Materials:**
   - Laminated question cards for each station.
   - Team slips or half-sheets for recording verified answers.
   - Answer keys for instructors to verify correctness.
   - Optional props or visual aids to enhance engagement (charts, diagrams, or manipulatives).

Instructions (detailed activity breakdown below) continued...

Facilitation Instructions

1. **Group Management:**
   - Assign students to pairs or small teams.
   - Allow teams to start at any station to reduce bottlenecks and encourage autonomy.
2. **Station Interaction:**
   - Students select a question card, collaborate to solve it, and record their answer on their team slip.
   - Encourage students to show their reasoning clearly; provide hints or clarifying questions but avoid giving answers.
   - Verify each answer promptly before students move to a new station.
   - Teams may revisit stations or select a new one once a problem is verified.
3. **Instructor Role:**
   - Circulate continuously, monitoring collaboration and engagement.
   - Ask guiding questions to promote deeper thinking or correct misconceptions.
   - Ensure teams follow the rules and document answers properly.
   - Keep the activity flowing by encouraging timely progress from station to station.

Rules and Expectations

1. Teams must collaborate fully; no switching partners mid-activity.
2. All work must be shown and explained before an answer is verified.
3. Direct copying of answers from other teams is prohibited.
4. Teams should communicate their reasoning with peers and instructors, but hints should remain guiding rather than directive.
5. Verified answers are the only ones that count toward scoring.

Instructions (detailed activity breakdown below) continued...

Scoring and Feedback

1. Each correctly verified answer earns points for the team.
2. Bonus points may be awarded for:
   - Clear demonstration of reasoning.
   - Correct use of units and notation.
   - Effective collaboration and communication.
3. Maintain a visible leaderboard to promote friendly competition and motivation.
4. Provide immediate verbal feedback at each station to reinforce correct approaches.

Wrap-Up and Reflection

1. After completion, collect all team slips and review common challenges observed during the activity.
2. Highlight effective problem-solving strategies and address common misconceptions.
3. Facilitate a brief discussion prompting students to reflect on:
   - Which types of questions were easiest or most challenging.
   - How collaboration improved understanding.
   - Key takeaways for upcoming assessments.
4. Summarize the conceptual links between all station topics and reinforce the practical application of the skills practiced.

Content (fully worked out problems & answers)

(1) A Brönsted-Lowry acid is defined as a substance that:

a. Donates a proton

b. Accepts a proton

c. Releases electrons

d. Accepts electrons

(2) What is the oxidation number of chlorine in NaCl?

a. +1

b. -1

c. 0

d. +2

(3) Which of the following reactions is a redox reaction?

a. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

b. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

c. $CaCO_3 \rightarrow CaO + CO_2$

d. $H_2O + SO_3 \rightarrow H_2SO_4$

(4) In the reaction $H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+$, the conjugate base is:

a. $H_2PO_4^-$

b. HPO_4^{2-}

c. H^+

d. None of the above

(5) For the equilibrium $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$, what happens when pressure increases?

a. The system shifts left because fewer gas moles are on the CO_2 side

b. The system shifts right because more gas moles are produced

c. No effect because solids and gases respond equally

d. Pressure changes only affect liquids, not gases

(6) What is the oxidation number of sulfur in SO_2?

a. +2

b. +4

c. +6

d. 0

(7) In the reaction $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, which is the Brönsted-Lowry acid?

a. HCl

b. H_2O

c. H_3O^+

d. Cl^-

(8) For the exothermic reaction $2SO_2 + O_2 \rightarrow 2SO_3$, increasing temperature will:

a. Increase SO_3 production

b. Shift equilibrium to left

c. Not affect equilibrium

d. Increase both forward and reverse rate equally

(9) Calcium hydroxide reacts with hydrochloric acid to form calcium chloride and water. Which statement best describes the acid-base behavior occurring in this reaction?

$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$

a. HCl donates protons to OH^- ions, forming water as Ca^{2+} pairs with Cl^-

b. $Ca(OH)_2$ donates protons to HCl, forming hydrogen gas

c. Both reactants accept protons, preventing water formation

d. Ca^{2+} acts as the proton donor while Cl^- acts as the proton acceptor

(10) For the reaction $R + S \rightleftharpoons RS$, which change could temporarily increase Q?

a. Adding RS

b. Adding R

c. Removing S

d. Diluting all species equally

(11) For the reaction $2A \rightleftharpoons B$, which expression is correct?

a. $K_{eq} = \frac{[B]}{[A]^2}$

b. $K_{eq} = \frac{[A]^2}{[B]}$

c. $K_{eq} = \frac{1}{[A][B]}$

d. $K_{eq} = [A][B]$

(12) Identify the reaction type:

$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

a. Combination

b. Decomposition

c. Exchange

d. Single replacement

(13) In a redox reaction, the oxidizing agent is the species that:

a. Loses electrons

b. Gains electrons

c. Is always neutral

d. Has the highest atomic mass

(14) For the endothermic reaction $N_2O_4 \rightarrow 2NO_2$, decreasing temperature favors:

a. Formation of NO_2

b. Formation of N_2O_4

c. Equal amounts of each

d. Faster formation of NO_2 only

(15) For the equilibrium $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, increasing pressure causes the reaction to shift:

a. Left, toward PCl_5

b. Right, toward PCl_3 and Cl_2

c. No shift because gases are unaffected by pressure

d. Toward whichever side has the higher total mass

(16) In the reaction $2Na + Cl_2 \rightarrow 2NaCl$, which is the oxidizing agent?

a. Na

b. Cl_2

c. NaCl

d. Both Na and Cl_2

(17) For the reaction $A + C \rightleftharpoons AC$ with $K_{eq} = 0.02$, which is true?

a. Reactants are strongly favored

b. Products are strongly favored

c. Equal amounts of A and AC form

d. Keq must be recalculated using solids only

(18) According to the Arrhenius definition, which statement correctly describes an acid and a base?

a. An acid produces H^+ in water, and a base produces OH^- in water

b. An acid donates a proton, and a base accepts a proton

c. An acid accepts an electron pair, and a base donates an electron pair

d. An acid and base both increase the concentration of ions equally

(19) What type of reaction occurs when methane burns in oxygen to produce carbon dioxide and water?

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

a. Combination

b. Decomposition

c. Combustion

d. Single displacement

(20) What type of chemical reaction occurs between sodium phosphate and calcium nitrate to form calcium phosphate and sodium nitrate?

$2Na_3PO_4 + 3Ca(NO_3)_2 \rightarrow Ca_3(PO_4)_2 + 6NaNO_3$

a. Decomposition

b. Exchange

c. Combination

d. Combustion

(21) In the reaction $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$, which change increases methanol production?

a. Increasing pressure because the product side has fewer gas moles

b. Decreasing pressure because the reactant side has more gas moles

c. Increasing volume of the container

d. Adding more inert gas at constant volume

(22) If $Q > K_{eq}$ for the reaction $F \rightleftharpoons G$, which direction will the system shift?

a. Toward reactants

b. Toward products

c. No shift

d. The reaction stops

(23) When sodium reacts with chlorine to produce sodium chloride, what type of reaction is this?
$2Na + Cl_2 \rightarrow 2NaCl$

a. Decomposition

b. Combustion

c. Exchange

d. Combination

(24) For the reaction $P \rightleftharpoons Q$, which condition must be true at equilibrium?

a. Rate forward = rate reverse

b. $[P] = [Q]$

c. $K_{eq} = 0$

d. All reactant is converted to product

(25) For the reaction A + heat → B, which statement is correct?

a. Adding heat increases [B]

b. Adding heat decreases [B]

c. Removing heat increases [B]

d. Heat has no effect on equilibrium

(26) The burning of benzene in air to form carbon dioxide and water is an example of what type of reaction?

$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$

a. Oxidation-reduction

b. Combustion

c. Decomposition

d. Neutralization

(27) Which element is being reduced in the reaction $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$?

a. Fe

b. O

c. C

d. CO_2

(28) What happens to an atom's oxidation number when it is oxidized?

a. It increases

b. It decreases

c. It stays the same

d. It becomes zero

(29) When ammonium chloride reacts with silver nitrate to form silver chloride and ammonium nitrate, the reaction is classified as:

$NH_4Cl + AgNO_3 \rightarrow AgCl + NH_4NO_3$

a. Combination

b. Exchange

c. Decomposition

d. Combustion

(30) What type of reaction occurs when potassium iodide reacts with lead(II) nitrate to form lead(II) iodide and potassium nitrate?

$2KI + Pb(NO_3)_2 \rightarrow 2KNO_3 + PbI_2$

a. Exchange

b. Combination

c. Decomposition

d. Combustion

(31) When hydrochloric acid reacts with sodium hydroxide to form sodium chloride and water, which statement best describes the acid-base process taking place?

$HCl + NaOH \rightarrow NaCl + H_2O$

a. HCl donates a proton to OH^-, forming water as Na^+ pairs with Cl^-

b. NaOH donates protons to HCl, forming hydrogen gas

c. Both reactants act as proton donors, preventing neutralization

d. Cl^- acts as the proton donor while Na^+ accepts the proton

(32) In the reaction $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$, which species acts as the Brönsted-Lowry base?

a. NH_3

b. H_2O

c. NH_4^+

d. OH^-

(33) Identify the reaction type:

$2H_2O_2 \rightarrow 2H_2O + O_2$

a. Combination

b. Combustion

c. Decomposition

d. Exchange

(34) Which pair represents a conjugate acid-base pair?

a. NH_3/NH_4^+

b. OH^-/O^{2-}

c. Na^+/Na

d. Cl^-/Cl_2

(35) Which of the following elements always has an oxidation number of +1 in compounds?

a. Sodium

b. Oxygen

c. Fluorine

d. Sulfur

(36) What is the oxidation number of manganese in $KMnO_4$?

a. +2

b. +4

c. +6

d. +7

(37) In a redox reaction, the reducing agent is the substance that:

a. Gains electrons

b. Loses electrons

c. Is neither oxidized nor reduced

d. Lowers its oxidation number

(38) For the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, decreasing pressure favors:

a. Formation of SO_3

b. Formation of SO_2 and O_2

c. No change in equilibrium

d. Formation of whichever side absorbs heat

(39) For the endothermic reaction heat$+H_2+I_2 \rightarrow 2HI$, decreasing temperature shifts equilibrium:

a. Toward products

b. Toward reactants

c. No shift

d. Depends only on pressure

(40) For the reaction $A + B \rightleftharpoons C$, if $K_{eq} = 12$, which statement is true?

a. Products are favored

b. Reactants are favored

c. Neither side is favored

d. The reaction goes to completion

(41) What type of reaction occurs when lithium sulfate reacts with barium nitrate to form barium sulfate and lithium nitrate?

$Li_2SO_4 + Ba(NO_3)_2 \rightarrow BaSO_4 + 2LiNO_3$

a. Combination

b. Decomposition

c. Exchange

d. Combustion

(42) For the reaction $2X \rightleftharpoons Y$, $K_{eq} = 1.0$. Which is true at equilibrium?

a. $[Y] = [X]$

b. $[Y] = [X]^2$

c. $[X] = 2[Y]$

d. K_{eq} cannot equal 1 for reversible reactions

(43) What kind of reaction is the burning of glucose in oxygen to form carbon dioxide and water?

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

a. Double displacement

b. Combustion

c. Decomposition

d. Combination

(44) A substance that gains electrons during a reaction is:

a. Oxidized

b. Reduced

c. Neutralized

d. Dissociated

(45) Which of the following represents a redox reaction?

a. $HCl + NaOH \rightarrow NaCl + H_2O$

b. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

c. $CaCO_3 \rightarrow CaO + CO_2$

d. $NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$

(46) For the exothermic decomposition $2NO_{(g)} \rightarrow N_{2(g)} + O_{2(g)} + \text{heat}$, adding heat will favor:

a. Formation of N_2 and O_2

b. Formation of NO

c. No change

d. Whichever side has fewer gas particles

(47) What type of reaction occurs when silver nitrate reacts with sodium chloride to produce silver chloride and sodium nitrate?

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

a. Combination

b. Decomposition

c. Exchange

d. Combustion

(48) For the reaction $H_2 + I_2 \rightleftharpoons 2HI$, which change increases Q?

a. Adding HI

b. Removing HI

c. Adding H_2

d. Removing heat

(49) What type of chemical reaction does ethanol undergo in air?

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

a. Neutralization

b. Single replacement

c. Combustion

d. Precipitation

(50) For the reaction $A + B + \text{heat} \rightarrow C$, which way does equilibrium shift when heat is added?

a. Toward products

b. Toward reactants

c. No change

d. It depends on pressure only

(51) For the exothermic reaction $X + Y \rightarrow Z + \text{heat}$, removing heat shifts equilibrium:

a. Toward reactants

b. Toward products

c. Causes no shift

d. Only affects rate, not equilibrium

(52) In the equilibrium $HF + H_2O \rightleftharpoons H_3O^+ + F^-$, which is the conjugate acid of F^-?

a. HF

b. H_3O^+

c. H_2O

d. F^- has no conjugate acid

(53) What type of reaction occurs when acetylene reacts with oxygen to produce CO_2 and H_2O?

$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$

a. Neutralization

b. Combustion

c. Combination

d. Decomposition

(54) For the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, increasing pressure shifts the equilibrium:

a. Toward products because fewer gas moles are present

b. Toward reactants because more gas moles are present

c. No shift because pressure does not affect gases

d. Toward whichever side has more volume

(55) In the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, which species is oxidized?

a. Cu^{2+}

b. Cu

c. Zn

d. H_2O

(56) Identify the type of reaction below:

$K_2SO_4 + Ba(NO_3)_2 \rightarrow 2KNO_3 + BaSO_4$

a. Exchange

b. Combination

c. Combustion

d. Decomposition

(57) For the reaction $A + C \rightleftharpoons AC$ with $K_{eq} = 0.02$, which is true?

a. Reactants are strongly favored

b. Products are strongly favored

c. Equal amounts of A and AC form

d. K_{eq} must be recalculated using solids only

(58) Which of the following compounds acts as an Arrhenius base when dissolved in water?

a. NaOH, because it increases the concentration of OH^- in solution

b. HCl, because it donates protons to other species

c. NH_3, because it donates an electron pair to H^+

d. CO_2, because it produces H^+ in water

(59) For $D \rightleftharpoons E$, which scenario decreases the value of the reaction quotient Q?

a. Adding D

b. Removing D

c. Adding E

d. Both b. and c.

(60) For the reaction $M \rightleftharpoons N$, if K_{eq} is extremely large, which is true?

a. Products dominate at equilibrium

b. Reactants dominate at equilibrium

c. $[M]$ must be greater than [N]

d. Q must equal zero

Checking for Understanding (questions & answers)

- How does adding a nonvolatile solute affect the freezing point of a solvent?
- Why does dissolving an ionic compound usually lower the freezing point more than a covalent one?
- What factors determine the magnitude of freezing point depression?
- Why does the identity (not just the amount) of the solute matter in colligative properties?
- How does the boiling point of a solution compare to that of the pure solvent?
- Why does adding solute increase the boiling point of a liquid?
- What physical property remains unchanged when a solution's freezing point is lowered?
- Why does temperature itself not change the amount of freezing point depression?
- Why do electrolytes and nonelectrolytes differ in their effects on colligative properties?
- How does a neutralization reaction differ from a precipitation or redox reaction?
- Why is acid–base neutralization classified as a double replacement reaction?
- What distinguishes a synthesis reaction from a decomposition reaction?
- How is a combustion reaction identified based on its reactants and products?
- Why does every combustion reaction of a hydrocarbon produce CO_2 and H_2O?
- How does a single replacement reaction differ from a double replacement reaction?
- Why must the activity of elements be considered when predicting single replacement outcomes?
- How can you identify the limiting reactant in a reaction from given amounts of reactants?
- Why does conservation of mass require balancing chemical equations?
- How can stoichiometric ratios be used to determine theoretical yield?
- Why do ionic compounds dissociate in water while molecular ones may not?
- How does solubility influence whether a precipitate forms in a double replacement reaction?
- Why is water often called the "universal solvent" in ionic and polar systems?
- How does polarity explain why oil and water do not mix?
- Why do stronger intermolecular forces lead to higher boiling points?
- How can temperature changes indicate whether a process is endothermic or exothermic?
- Why is combustion considered an exothermic oxidation process?
- What evidence suggests that a chemical reaction, not a physical change, has occurred?

Activity 2

Collaborative Learning Techniques	Jigsaw
Learning Strategy	Predict Test Questions
Time	12:05—12:30
Materials	Pencil, eraser, calculator, periodic table
Lecture/Textbook/Old Plans/Slide References	Chapter 10: Acid and Bases

Definition

1. **Jigsaw:** Jigsaws, when used properly, make the group as a whole dependent upon all of them in subgroups. Each group provides a peice of the puzzle. Group members are broken into smaller groups. Each small group works on some aspect of the same problem, question, or issue. They then share their part of the puzzle with the large group.

2. **Predict Test Questions:** Put students in groups of 2 or 3 and assign them to write a test question for a specific topic, ensuring that all topics have been convered. Ask students to write their questions on the board or on an overhead for discussion (would the professor ask the question? What is the answer? etc.). Students will have the benefit of learning to think like the teacher and they'll be able to see additional questions that other students have writen.

Instructions (detailed activity breakdown below)

1. At each station, place a set of laminated question cards (multiple choice and true/false). Each card includes:
   - A unique question number and topic label.
   - The possible answer choices (A–E or T/F).
   - A mapping line showing which station each answer leads to (to simulate "pathways" through the escape room).
2. Provide each pair of students with a **team slip** (half-sheet) labeled with their team name and space to record up to 3 verified answers.
3. **Facilitation:**
   - Pairs may begin at **any station of their choice**.
   - Each pair selects one question card, discusses the problem, and writes their chosen answer **on the back of the card**.
   - Students then show their work and reasoning to the instructor or SI leader for quick verification.
   - If the answer is correct, they record that problem on their team slip (problem number and answer) and may proceed to any other station of their choice.
   - Pairs are encouraged to move freely between stations, aiming to correctly solve at least **three problems** during the activity period.
   - After completing at least three problems, teams submit their slips to the instructor for scoring.
4. **Scoring:**
   - Each correctly solved problem = **1 point**.
   - Teams with the highest total score win or "escape" first.
   - Bonus points may be awarded for clear reasoning or teamwork.
5. **Rules:**
   - Teams must work collaboratively—no switching partners mid-activity.
   - All work must be shown before a response is accepted.
   - Only verified answers written on the team slip count toward the final score.
   - Communication between pairs is encouraged, but direct sharing of answers is not permitted.
6. **Wrap-Up:**
   - After all slips are collected, review several challenging questions as a class.
   - Discuss common reasoning errors and strategies used by top-scoring pairs.
   - Reinforce conceptual links between stations (e.g., gas behavior and intermolecular forces).

Content (fully worked out problems & answers)

(1) Which solution has the highest $[OH^-]$?

a. pH = 6.10

b. pH = 8.23

c. pH = 9.50

d. pH = 3.33

(2) Which substance is a strong base?

a. NH_3

b. $Ca(OH)_2$

c. HF

d. HNO_2

(3) Which of the following statements about salt hydrolysis is correct?

a. Only salts of strong acids hydrolyze

b. Only salts of strong bases hydrolyze

c. Salts of weak acids or weak bases hydrolyze

d. No salts hydrolyze in water

(4) Which salt forms a basic solution upon hydrolysis?

a. KClO

b. NH_4Br

c. $NaNO_3$

d. LiCl

(5) Which describes a neutral solution?

a. $[H_3O^+] > [OH^-]$

b. $[H_3O^+] < [OH^-]$

c. $[H_3O^+] = [OH^-]$

d. $[H_3O^+] = 0$

(6) A solution has $[H_3O^+] = 3.20 \times 10^{-6}$ M. Its pH is closest to:

a. 4.49

b. 5.49

c. 6.49

d. 7.49

(7) Which salt will produce a basic solution upon dissolution due to its parent species?

a. NaF

b. NH_4Br

c. KCl

d. $Ca(NO_3)_2$

(8) In the reaction $HCOOH + H_2O \rightleftharpoons HCOO^- + H_3O^+$, HCOOH is acting as a(n):

a. Base

b. Acid

c. Salt

d. Neutral compound

(9) Which of the following solutions is most basic?

a. pH = 9.00

b. pH = 11.00

c. pH = 7.00

d. pH = 5.00

(10) Which of the following best explains why pKa values are useful?

a. They show how fast an acid reacts

b. They show the strength of an acid using a log scale

c. They show solubility of an acid

d. They show the molecular mass of an acid

(11) Which term describes an acid that ionizes partially in water?

a. Strong acid

b. Weak acid

c. Strong base

d. Neutral compound

(12) Which description correctly identifies the parent acid of the salt sodium acetate (NaOAc)?

a. HCl

b. CH_3COOH

c. HNO_3

d. HF

(13) Which species is the conjugate acid of NH_3?

a. NH_2^-

b. NH_4^+

c. N^{3-}

d. NH_2OH

(14) In the stepwise dissociation of H_2SO_4, which statement is correct?

a. Both protons dissociate completely

b. The first proton dissociates strongly, the second weakly

c. Neither proton dissociates

d. The second proton dissociates more strongly than the first

(15) For the equilibrium $HA + H_2O \rightleftharpoons H_3O^+ + A^-$, which expression represents K_a?

a. $\dfrac{[HA]}{[H_3O^+][A^-]}$

b. $\dfrac{[H_3O^+][A^-]}{[HA]}$

c. $[H_3O^+][A^-][HA]$

d. $\dfrac{1}{[HA]}$

(16) In the reaction $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$, which species acts as the Brönsted-Lowry base?

a. NH_3

b. H_2O

c. NH_4^+

d. OH^-

(17) Which species is amphiprotic?

a. OH^-

b. HSO_4^-

c. Cl^-

d. NH_4^+

(18) A solution has pH = 5.00. What is the hydronium concentration?

a. 1.00×10^{-3} M

b. 1.00×10^{-5} M

c. 1.00×10^{-7} M

d. 1.00×10^{-9} M

(19) If a salt's anion is the conjugate base of a weak acid, what will be the pH of its aqueous solution?

a. Acidic

b. Basic

c. Neutral

d. Impossible to determine

(20) A weak acid has Ka = 3.00×10^{-7}. Which pKa is closest?

a. 3.96

b. 4.22

c. 5.44

d. 6.80

(21) A solution has a hydronium concentration of 1.0×10^{-4} M. What is its pH?

a. 2

b. 3

c. 4

d. 5

(22) Which statement correctly distinguishes dissociation from ionization?

a. Dissociation involves breaking covalent bonds to form new ions, while ionization separates pre-existing ions in an ionic lattice.

b. Dissociation separates pre-existing ions from an ionic compound without altering their identity or overall charge, while ionization forms ions from a molecular substance through a chemical change.

c. Dissociation always produces H_3O^+, while ionization never does.

d. Dissociation only occurs in strong acids, while ionization only occurs in weak acids.

(23) A weak base has $[OH^-] = 1.00 \times 10^{-5}$ M. What is the pOH?

a. 3.00

b. 5.00

c. 7.00

d. 9.00

(24) For the equilibrium $HF + H_2O \rightleftharpoons H_3O^+ + F^-$, which species is the conjugate base?

a. HF

b. H_3O^+

c. F^-

d. H_2O

(25) A base that ionizes only slightly in water is classified as a:

a. Strong base

b. Weak base

c. Neutral molecule

d. Strong acid

(26) The parent acid of NO_2^- is:

a. HNO_3

b. HNO_2

c. HNO

d. HNO_4

(27) A solution has pOH = 3.60. Determine the pH, $[H_3O^+]$, and $[OH^-]$ of the solution. Ensure concentrations are in molarity and pH/pOH to two decimals.

(28) Calcium hydroxide reacts with hydrochloric acid to form calcium chloride and water. Which statement best describes the acid-base behavior occurring in this reaction?

$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$

a. HCl donates protons to OH^- ions, forming water as Ca^{2+} pairs with Cl^-

b. $Ca(OH)_2$ donates protons to HCl, forming hydrogen gas

c. Both reactants accept protons, preventing water formation

d. Ca^{2+} acts as the proton donor while Cl^- acts as the proton acceptor

(29) Which statement correctly describes a strong base?

a. Completely dissociates in water

b. Partially dissociates in water

c. Does not react with water

d. Cannot accept protons

(30) A solution with pOH = 3.00 has what $[OH^-]$?

a. 1.00×10^{-3} M

b. 1.00×10^{-7} M

c. 1.00×10^{-11} M

d. 1.00×10^{-14} M

(31) Which statement best describes the logarithmic nature of pKa?

a. A decrease of 1 pKa means the acid is ten times stronger

b. A decrease of 1 pKa means the acid is half as strong

c. A decrease of 1 pKa means no measurable change

d. A decrease of 1 pKa means the acid becomes more basic

(32) Which of the following is a triprotic acid?

a. HCl

b. H_2CO_3

c. H_3PO_4

d. HF

(33) Which is the conjugate acid of CO_3^{2-}?

a. CO_2

b. HCO_3^-

c. H_2CO_3

d. CO_3H_2

(34) Which salt will undergo acidic hydrolysis when dissolved in water?

a. KNO_3

b. NH_4Cl

c. $NaClO_3$

d. $CaCl_2$

(35) A solution has pH = 11.00 What is the hydroxide concentration?

a. 1.00×10^{-3} M

b. 1.00×10^{-11} M

c. 1.00×10^{-7} M

d. 1.00×10^{-14} M

(36) A Brönsted-Lowry acid is defined as a substance that:

a. Donates a proton

b. Accepts a proton

c. Releases electrons

d. Accepts electrons

(37) A weak acid has pKa = 4.25 and [HA] = 0.10 M. If $[H_3O^+] = 5.6 \times 10^{-5}$ M, calculate pH, pOH, and $[OH^-]$. Ensure concentrations are in molarity and pH/pOH to two decimals.

(38) In the reaction $H_2PO_4^- + H_2O \rightleftharpoons HPO_4^{2-} + H_3O^+$, the conjugate base is:

a. $H_2PO_4^-$

b. HPO_4^{2-}

c. H_3O^+

d. None of the above

(39) Which of the following best defines ionization for a weak acid in water?

a. Complete separation into ions

b. Partial separation into ions

c. Reaction with a base to form a salt

d. Combination with water molecules without forming ions

(40) A weak acid has pKa = 3.97. Its solution has $[H_3O^+] = 8.00 \times 10^{-5}$ M. Determine pH, pOH, and $[OH^-]$. Ensure concentrations are in molarity and pH/pOH to two decimals.

(41) Which pair represents a conjugate acid-base pair?

a. NH_3/NH_4^+

b. OH^-/O^{2-}

c. Na^+/Na

d. Cl^-/Cl_2

(42) Which describes the relationship between Ka and acid strength?

a. Larger Ka means stronger acid

b. Smaller Ka means stronger acid

c. Ka does not relate to strength

d. Ka only applies to bases

(43) If the pH of a solution increases from 8.00 to 10.00, the $[H^+]$ changes by a factor of:

a. 2x

b. 10x

c. 50x

d. 100x

(44) A strong base solution has $[OH^-] = 6.08 \times 10^{-4}$ M. Calculate pOH, pH, and $[H_3O^+]$ to two decimal places, with concentrations in molarity.

(45) A solution with pH = 2.00 has a hydronium concentration that is how many times larger than a solution with pH = 4.00?

a. 2x

b. 10x

c. 100x

d. 1000x

(46) If an acid has $K_a = 1.00 \times 10^{-5}$, which of the following is closest to its pKa?

a. 2.20

b. 3.50

c. 5.33

d. 7.25

(47) A solution has pH = 5.83. Find $[H_3O^+]$, pOH, and $[OH^-]$. Concentrations in M, pH/pOH to two decimal places.

(48) Which statement correctly defines ionization for a base in water?

a. Forming OH^- by reacting chemically with water rather than releasing pre-existing ions

b. Separating into pre-existing ions without chemical change

c. Producing H_3O^+ by transferring a proton to water

d. Splitting into neutral molecules without forming ions

(49) Which species will act as a base according to Brönsted-Lowry?

a. H_3O^+

b. NH_3

c. HCl

d. HNO_3

(50) A solution has $[H_3O^+] = 2.22 \times 10^{-6}$ M. Calculate the pH, pOH, and $[OH^-]$.

(51) Which of the following is a strong acid?

a. HF

b. HCl

c. HCN

d. CH_3COOH

(52) A solution has $[H_3O^+] = 4.58 \times 10^{-5}$ M. Calculate the pH, pOH, and $[OH^-]$. Express concentrations in molarity and pH/pOH to two decimal places.

(53) Which salt produces an acidic solution?

a. NaCl

b. KCN

c. NH_4NO_3

d. NaOAc

(54) Which salt is formed from a strong acid and strong base and therefore produces a neutral solution?

a. NaCl

b. NH_4Cl

c. NaOAc

d. KCN

(55) When hydrochloric acid reacts with sodium hydroxide to form sodium chloride and water, which statement best describes the acid-base process taking place?

$HCl + NaOH \rightarrow NaCl + H_2O$

a. HCl donates a proton to OH^-, forming water as Na^+ pairs with Cl^-

b. NaOH donates protons to HCl, forming hydrogen gas

c. Both reactants act as proton donors, preventing neutralization

d. Cl^- acts as the proton donor while Na^+ accepts the proton

(56) A weak acid HA has a pKa of 4.70. Which of the following pH values will favor the deprotonated form, A^-?

a. 2.00

b. 3.00

c. 4.70

d. 7.00

(57) According to the Arrhenius definition, which statement correctly describes an acid and a base?

a. An acid produces H^+ in water, and a base produces OH^- in water

b. An acid donates a proton, and a base accepts a proton

c. An acid accepts an electron pair, and a base donates an electron pair

d. An acid and base both increase the concentration of ions equally

(58) In the equilibrium $HF + H_2O \rightleftharpoons H_3O^+ + F^-$, which is the conjugate acid of F^-?

a. HF

b. H_3O^+

c. H_2O

d. F^- has no conjugate acid

(59) Which acid has the strongest conjugate base?

a. HCl

b. HNO_3

c. HF

d. CH_3COOH

(60) Which statement is true about conjugate acid-base pairs?

a. They differ by two protons

b. They differ by one proton

c. They differ by one electron

d. They differ by two electrons

Checking for Understanding (questions)

1. How do you distinguish between dissociation and ionization in aqueous solutions? Give examples of each.
2. How is the hydronium concentration $[H_3O^+]$ related to pH? How do you calculate one from the other?
3. How do you determine pOH and $[OH^-]$ from a known pH?
4. What is the relationship between Ka and acid strength? How can you estimate which species is more acidic?
5. How do you calculate percent ionization for a weak acid given $[H_3O^+]$ and initial concentration?
6. How do pKa values inform which form (protonated or deprotonated) predominates at a given pH?
7. How do you identify the conjugate acid or base in a given reaction?
8. How do you predict the effect of a strong acid vs. weak acid on pH?
9. How does a diprotic or triprotic acid behave differently in terms of sequential ionization?
10. How do you determine whether a salt solution will be acidic, basic, or neutral based on the strengths of its parent acid and base?
11. What is salt hydrolysis, and how does it affect the pH of the solution?
12. How do you check that your pH, pOH, $[H_3O^+]$, and $[OH^-]$ calculations are consistent?
13. Why is it important to include units for concentration (M) and round pH/pOH to two decimal places?
14. How do strong acids and strong bases differ from weak acids and weak bases in terms of dissociation/ionization?
15. How can you predict the direction of proton transfer in a Brønsted-Lowry acid-base reaction?

Closer

Collaborative Learning Techniques	Group Survey
Learning Strategy	Brain dump
Time	12:30—12:32
Materials	Index cards, writing utensils
Lecture/Textbook/Old Plans/Slide References	Chapters 1-10

Definitions

1. **Group survey:** Each group member is surveyed to discover their position on an issue, problem or topic. This process ensures that each member of the group is allowed to offer or state their point of view.

2. **Brain dump:** This is a great closer for students to sum up their knowledge on the topic(s) from the session. Have the students take out a blank piece of paper /Google Doc and write/type("dump") everything they know relevant from the discussion that day. The SI Leader can look at their work to assess the students' understanding of the course material and have them compare it to each other. It is important to at least discuss with the students so the SI Leader knows the students' comfort levels with the material.

Instructions (please provide a DETAILED activity breakdown below (and/or attached

1. Students are put into small groups and are asked to write down their confidence on a single notecard without names.
2. The cards are evidence of the group's confidence in the content.

Content (fully worked out problems & answers)

- No new worked problems; brain dump.

Checking for Understanding (questions & answers)

- Review confidence levels
- Interpret the variety of confidence and their distribution in groups.
- Get feedback for what they like and don't like
- Check in if the time is suitable for the students

Plan Checklist

- ☐ Variety of Collaborative Learning Techniques and Learning Strategies
- ☐ Chapter/slide# where information is coming from
- ☐ Included timestamps (and buffer time as needed)
- ☐ Details on how leader will break up students
- ☐ Checking for Understanding questions
- ☐ Fully worked out problems and examples for each activity
- ☐ Necessary links required for online implementation (if applicable)
- ☐ Source material correctly cited (if applicable)
- ☐ Plan is uploaded to Teams